Logarithms

1 Rewrite each of the following equations in logarithmic form:

a $5^2 = 25$

b $3^x = 81$

c $3^1 = 3$

d $2^0 = 1$

e $4.4^0 = 1$

f $25^{1.5} = 125$

g $4^{\frac{5}{2}} = 32$

h $4^{-2} = 0.0625$

i $4^{-3} = \dfrac{1}{64}$

j $x^{1.5} = 64$

2 Rewrite each of the following equations in logarithmic form:

a $10^2 = 100$

b $10^{-2} = \dfrac{1}{100}$

c $10^{\frac{1}{2}} = \sqrt{10}$

d $10^{-\frac{1}{2}} = \dfrac{1}{\sqrt{10}}$

3 Rewrite each of the following equations in exponential form:

a $\log_4 16 = 2$

b $\log_5 5 = 1$

c $\log_8 1 = 0$

d $\log_2 0.125 = -3$

e $\log_3 \dfrac{1}{3} = -1$

f $\log_{5.8} 33.64 = 2$

g $\log_{\frac{1}{3}} 9 = -2$

h $\log_x 32 = 5$

i $\log_8 x = 6$

4 Rewrite each of the following equations in exponential form:

a $\log_{10} 10\,000 = 4$

b $\log_{10} \left(\sqrt{10}\right) = \dfrac{1}{2}$

c $\log_{10} \left(\dfrac{1}{10\,000}\right) = -4$

d $\log_{10} \left(\dfrac{1}{\sqrt{10}}\right) = -\dfrac{1}{2}$

5 True or false: The equation $4\,(1.09)^x = 20$ is equivalent to $x = \log_{1.09} 5$.

6 The logarithm of what number is equal to zero?

7 Evaluate the following logarithmic expressions:

a $\log_2 16$

b $\log_8 \left(\dfrac{1}{64}\right)$

c $\log_5 0.2$

d $\log_4 1$

e $\log_{36} 6$

f $\log_2 \left(\dfrac{1}{4}\right)$

g $\log_{10} 0.1$

h $\log_7 \sqrt[3]{7}$

i $2^{\log_2 3}$

j $\log_2 \sqrt[4]{2}$

k $\log_3 3$

l $\log_{16} \sqrt{2}$

m $\log_2 \left(\sqrt[3]{\dfrac{1}{16}}\right)$

n $\log_5 125^{\frac{5}{4}}$

8 Evaluate $\log_{10} 5.16$, correct to three decimal places.

9 In order to estimate $\log_{10} 90$ without a calc[redacted]r:

a Evaluate $\log_{10} 10$.

b Evaluate $\log_{10} 100$.

c Between which two consecutive integers would you expect $\log_{10} 90$ to lie?

10 Consider the expression $\log 8892$.

a Between which two consecutive integers would you expect $\log 8892$ to lie?

b Find the value of $\log 8892$ correct to four decimal places.

11 Consider the function $f(x) = \log_{10}(-6x)$. Determine whether each of the following is defined. If so, evaluate the expression correct to two decimal places.

a $f(3)$

b $f\left(\frac{1}{3}\right)$

c $f\left(-\frac{1}{3}\right)$

d $f(-3)$

12 Consider the function $f(x) = \log_2(x-2)$. Evaluate $f(10)$.

13 Consider the function $f(x) = \log_2 x + 3$. Evaluate:

a $f(8)$

b $f\left(\frac{1}{8}\right)$

c $f(32)$

d $f\left(\frac{1}{64}\right)$

14 True or false: If $5^{x-6} = 2^{4x+3}$, then $x - 6 = \log_5 2^{4x+3}$.

15 Consider the function $f(x) = \log_{10}(4x)$. Solve for the value of x for which $f(x) = 2$.

16 Consider the function $f(x) = \log_{10}(3x+9)$. Evaluate $f(4)$ to two decimal places.

17 Consider the function $f(x) = \log_4(kx) + 1$. Solve for the value of k for which $f(4) = 3$.

18 Consider the function $f(x) = 4\log_{\frac{1}{4}} x$.

a Evaluate $f\left(\frac{1}{4}\right)$.

b Solve the value for x for which $f(x) = 0$.

19 For each of the following equations:

i Rewrite the equation in logarithmic form.

ii Approximate the value of x to two decimal places.

20 Given that $a > 1$, fully simplify the following expressions:

a $\log_a\left(\frac{1}{a}\right)$

b $\log_a\left(a^9\right)$

c $\log_a\left(\frac{1}{a^2}\right)$

d $\log_a\left(\sqrt{a}\right)$

e $\log_a\left(\frac{1}{\sqrt{a}}\right)$